

# Chuyang (Cyrus) Zhang

Software Engineer | Weather AI & Data

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Greater Seattle Area, WA, US, 98004

Software Engineer with experience building weather AI systems, scientific data pipelines, and satellite intelligence workflows at RWE's AI Lab. Worked alongside scientists to develop model-ready GOES/SMAP datasets, data quality validation, and Aurora training data and ensemble tooling on on-prem GPU/HPC infrastructure. Built AWS based geospatial pipelines and agentic automation for renewable energy project screening, with broader work spanning energy market modeling and trading research.

## Work Experience

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### Software Engineer, Commercial AI Lab

Jan 2025 – Jun 2026

*RWE Supply & Trading GmbH | Bellevue, WA*

- Built an AWS-based geospatial intelligence pipeline and agentic screening system endorsed by RWE's VP of Global Transmission, integrating Vantor/Maxar satellite imagery discovery, ordering, and delivery APIs, OpenStreetMap grid topology, CRS-aware area-of-interest handling, and Claude Vision to screen ~200 candidate substations for renewable energy and battery storage development, increasing screening throughput from 2-3 sites per day to hundreds in hours.
- Engineered cloud-native data pipelines that transformed raw scientific datasets spanning multiple grid resolutions, including a 2015–2025 NASA SMAP L4 soil moisture archive and GOES radiance datasets, into model-ready assets using Python, xarray, xESMF, Zarr v3, Icechunk, and Earthmover ArrayLake.
- Developed and shipped a CI-ready GOES reprojection validation system using mutual information/NMI scoring, multiprocessing, resumable state, thresholding, and joint histogram diagnostics, identifying data failures during satellite sensor calibration periods before they entered downstream model ingestion and training workflows.
- Built model-training data pipelines for an Aurora-based lightweight decoder model, aligning 1024-dimensional weather latents with ERCOT load targets and 2015-2025 European day-ahead electricity prices, including leakage-safe splits, hourly/lagged targets, gas/carbon/renewable auxiliaries, normalized Parquet artifacts, and reusable normalizers.
- Implemented and merged Aurora bred-vector ensemble tooling for PMF (Probabilistic Medium-range Forecasting), using ECMWF initial condition/Z500 perturbation workflows, spherical harmonic/correlated noise experiments, 24-hour breeding cycles, 6-hour rescaling, positive/negative members, and 240-hour autoregressive rollouts to enable up to 1000 ensemble forecasts on internal GPU/HPC infrastructure.
- Built and benchmarked self-hosted LLM alternatives to frontier APIs for production n8n agentic workflows, integrating satellite and news agents with Qwen/Nemotron models on internal H100 infrastructure and building evaluation workflows to diagnose reasoning streams, tool calling, token budgets, structured outputs, latency, and model routing tradeoffs.
- Deployed and adapted a LangGraph-based TradingAgents framework on self-hosted LLM infrastructure, running a 13+ agent equity analysis pipeline with full state JSON logs and designing an energy commodity extension using weather, load, renewables, gas, power market, and news signals.

### AI/ML Data Engineer

Apr 2022 - Apr 2023

*Prescriptive Health, Inc. | Redmond, WA*

- Built incremental PySpark ingestion and Azure Data Factory ETL pipelines for PHI/HIPAA data, reducing cluster compute load by 50% and supporting onboarding of 1000+ clients while improving secure cross-team data access.
- Migrated pipelines to Delta Lake/Databricks and supported ML serving foundations with Feature Store, Docker, Azure Model Registry, AKS, and Terraform, improving query efficiency by 80% while reducing data footprint/cloud storage cost and preparing real-time inference workflows.

## Core Skills

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Python, PyTorch, Machine Learning, Deep Learning, LLMs, Agentic AI, LangGraph, LangChain, Multi-Agent Systems, Ollama, Claude/OpenAI APIs, Generative AI, Computer Vision, xarray, Zarr/Icechunk, NumPy/Pandas, PySpark, Spark, Databricks, Delta Lake, SQL, Distributed Systems, AWS, Docker, Kubernetes, LLM Evaluation, Prefect, Slurm, Linux, CUDA, Git, CI/CD, MLOps

## Education

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**CORNELL UNIVERSITY - Master** Applied Statistics

Sep 2020 - Dec 2021

**OHIO STATE UNIVERSITY - Bachelor of Science** Mathematics, Statistics

Sep 2016 - May 2020